

Instructions. (15 points) Sections 16.7, 16.8, 16.9. Show relevant work for full credit.

- (3^{pts}) **1.** Compute $\iint_S z \, dS$ where S is the cylindrical surface $x^2 + y^2 = 1$, $0 \leq z \leq 2$.

Solution:

Parametrize the cylinder by $\mathbf{r}(\theta, z) = \langle \cos \theta, \sin \theta, z \rangle$ for $0 \leq \theta \leq 2\pi$, $0 \leq z \leq 2$. Then

$$\mathbf{r}_\theta = \langle -\sin \theta, \cos \theta, 0 \rangle, \quad \mathbf{r}_z = \langle 0, 0, 1 \rangle, \quad \mathbf{r}_\theta \times \mathbf{r}_z = \langle \cos \theta, \sin \theta, 0 \rangle,$$

so $|\mathbf{r}_\theta \times \mathbf{r}_z| = 1$. Therefore

$$\iint_S z \, dS = \int_0^{2\pi} \int_0^2 z \cdot 1 \, dz \, d\theta = 2\pi \cdot 2 = \boxed{4\pi}.$$

- (3^{pts}) **2.** Compute $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F}(x, y, z) = \langle 0, 0, z \rangle$ and S is the part of the paraboloid $z = 4 - x^2 - y^2$ lying above the xy -plane, oriented *upward*.

Solution:

The surface is the graph $z = g(x, y) = 4 - x^2 - y^2$ over the disk $D : x^2 + y^2 \leq 4$. For an upward-oriented graph,

$$d\mathbf{S} = \langle -g_x, -g_y, 1 \rangle \, dA = \langle 2x, 2y, 1 \rangle \, dA, \quad \mathbf{F} \cdot d\mathbf{S} = z \, dA = (4 - x^2 - y^2) \, dA.$$

In polar:

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \int_0^{2\pi} \int_0^2 (4 - r^2) r \, dr \, d\theta = 2\pi(8 - 4) = \boxed{8\pi}.$$

- (3^{pts}) **3.** Let $\mathbf{F}(x, y, z) = \langle z^2, 2x, -y^3 \rangle$ and let C be the unit circle $x^2 + y^2 = 1$ in the plane $z = 0$, oriented **clockwise** when viewed from above. Use Stokes' Theorem to compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$.

Solution:

Choose a surface and orient it. Take S to be the unit disk $\{x^2 + y^2 \leq 1, z = 0\}$. Because C is traversed **clockwise** viewed from above, the right-hand rule forces the surface normal to point **downward**: $d\mathbf{S} = \langle 0, 0, -1 \rangle dA$.

Compute the curl.

$$\nabla \times \mathbf{F} = \langle (-y^3)_y - (2x)_z, (z^2)_z - (-y^3)_x, (2x)_x - (z^2)_y \rangle = \langle -3y^2, 2z, 2 \rangle.$$

On S ($z = 0$): $\nabla \times \mathbf{F} = \langle -3y^2, 0, 2 \rangle$.

Apply Stokes'.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \iint_S (2)(-1) dA = -2 \cdot \text{Area}(S) = \boxed{-2\pi}.$$

- (3^{pts}) **4.** Let $\mathbf{F}(x, y, z) = \langle xz^2, y^3, x^2z \rangle$ and let S be the unit sphere $x^2 + y^2 + z^2 = 1$ with outward orientation. Use the Divergence Theorem to compute $\iint_S \mathbf{F} \cdot d\mathbf{S}$.

Solution:

$\nabla \cdot \mathbf{F} = z^2 + 3y^2 + x^2$. With E the closed unit ball,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E (x^2 + 3y^2 + z^2) dV.$$

Symmetry shortcut. By the symmetry of the unit ball, $\iiint_E x^2 dV = \iiint_E y^2 dV = \iiint_E z^2 dV$. Their sum equals

$$\iiint_E (x^2 + y^2 + z^2) dV = \iiint_E \rho^2 dV = \int_0^{2\pi} \int_0^\pi \int_0^1 \rho^4 \sin \phi d\rho d\phi d\theta = 2\pi \cdot 2 \cdot \frac{1}{5} = \frac{4\pi}{5},$$

so each integral equals $\frac{1}{3} \cdot \frac{4\pi}{5} = \frac{4\pi}{15}$.

Therefore

$$\iiint_E (x^2 + 3y^2 + z^2) dV = (1 + 3 + 1) \cdot \frac{4\pi}{15} = \boxed{\frac{4\pi}{3}}.$$

(3^{pts}) 5. For each of the following, identify the single *best* method for evaluation: **direct parametrization**, **FTLI**, **Green's Theorem**, **Stokes' Theorem**, or the **Divergence Theorem**. Give a one-sentence reason. (You do not need to evaluate any of them.)

(a) $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = \langle xy, e^{x^2}, z \rangle$ and C is the triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ traversed once.

(b) $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F} = \langle x + e^y, y + \sin z, z \rangle$ and S is the boundary of the cube $[0, 1]^3$ with outward orientation.

(c) $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F} = \langle 2x, 2y, 2z \rangle$ and C is any smooth path from $(0, 0, 0)$ to $(1, 2, 3)$.

Solution:

(a) **Stokes' Theorem.** C is a closed curve in 3D, and direct parametrization is awkward (three segments, integrand contains e^{x^2} which has no elementary antiderivative). Replace with the curl flux through the flat triangular surface S whose boundary is C .

(b) **Divergence Theorem.** S is a closed surface bounding a solid, and $\nabla \cdot \mathbf{F} = 1 + 1 + 1 = 3$ is simple, so $\iiint_E 3 dV = 3 \cdot \text{Vol} = 3$ is far easier than computing flux through six faces.

(c) **FTLI.** $\mathbf{F} = \nabla(x^2 + y^2 + z^2)$ is conservative, so $\int_C \mathbf{F} \cdot d\mathbf{r} = f(1, 2, 3) - f(0, 0, 0) = 14$ no matter what C is.